

Interpretation of solute profile dynamics in irrigated soils

II. Convective-dispersive approach

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Summary. The simple field-oriented convective-dispersive equation based model of Rose et al. (1982a and b) has potential to be used to monitor changes in soil chloride levels on commercial farms. As yet applications of the model have only been reported for detailed studies conducted on coarse textured soils. Here the model was applied to a wider range of soil types using input data likely to be available on commercial farms.

The model gave good predictions of measured soil chloride profiles at five sites, which included two spray-irrigated Vertisols. To provide accurate predictions in these two Vertisols, infiltration of rainfall into macropores had to be accounted for (by an approximate means). Values of parameters in the model (including the solute dispersivity coefficient) at these five sites were within the range found in previous studies, even for the Vertisols. Variation in the solute dispersivity coefficient within the extreme values that had been found in this and previous studies, did not greatly affect predicted soil chloride profiles.

At another three sites, all flood irrigated Vertisols, predicted soil chloride levels were lower than measured values, regardless of values chosen for model parameters. The effect of depth variation in field capacity and wilting point moisture contents on chloride movement was investigated at these three sites as a possible cause of the inaccurate predictions, and found to have a negligible effect on predicted profiles. Differences between measured and predicted soil chloride levels were attributed to infiltration of irrigation water into cracks and, at one site, to the effects of high watertables. These processes were not represented in the model, explaining the failure in this context. Application of the PEAK model provided useful insights into the root-zone hydrology of all soils studied.

salinity or solute movement to groundwater, so that deleterious consequences of irrigation for agriculture or the environment can be avoided. Agricultural or environmental advisory agencies are often called upon to perform this assessment, and so require models that can be routinely applied over a wide range of soil types with varying detail of irrigation management or soils data. Simple mass balance models (e.g. USSL 1954) may adequately describe the effect of irrigation on soil salinity in many situations. However, these models require stable management through time and provide little detail of the depth distribution or short-term temporal variation of solutes within the root-zone. Where irrigation management varies, or short-term predictions are required, more detailed solute transport models are needed.

There are many detailed models of solute transport available, but generally only the simpler and less mechanistic are suitable for field applications (Addiscott and Wagenet 1985), because of practical limitations in gathering realistic solute transport parameters for field soils (Rose et al. 1982c; Jury 1982). This problem is further aggravated when working on commercial farms where difficulties can be encountered even in obtaining reliable irrigation data, limiting the value of detailed models.

Different approaches have been taken in developing simple solute transport models for field management applications. The widely applied models of Burns (1974) and Addiscott (1977) describe solute movement between arbitrarily defined finite soil layers. These models have been used to predict field measured chloride profiles with success in many studies (e.g. Nicholls et al. 1982; Khanif et al. 1984), although they are less successful when the assumptions regarding solute movement between the layers are not met (Cameron and Wild 1982; Barraclough et al. 1983; Madan Kumar and Oswal 1984). In contrast, Rose et al. (1982; a, b and c) used water balance accounting principles in combination with an analytical solution of the convective-dispersion equation to calculate solute concentrations within soil profiles at any given depth and time. Applications of this model (subsequently referred to as the PEAK model) have so far only been reported for

Solute transport models provide a means of assessing the effect of different irrigation management practices on soil

Table 1. Details of previously reported applications of the PEAK model; texture of soils studied and values of model parameters determined

Study	Soil texture	ε (mm)	j	D_0 (mm ² day ⁻¹)
Rose et al. (1982a, b and c)	loamy sand	120	0	100
Cameron and Wild (1982)	silty loam	30 (10–100) ^b	ng ^a	10 (10–100)
Barry et al. (1985)	silty clay	100 (90–110)	1	100
De Smedt and Wierenga (1978) ^c	silty clay loam	5 (5–20)	ng	300 (0–300)

^a ng = details not given^b range of values investigated^c laboratory simulation using the equation $C = C_0 \frac{1}{2} [\text{erfc}(A)]$, cf. Eq. (5)

light textured soils (Table 1) studied under controlled experimental conditions. It gave accurate predictions in these studies, and may be suitable for wider use.

There are possible limitations to wider application of the PEAK model. Firstly, a value of the field solute dispersivity coefficient (ε) is required for the model. Values of ε have been derived from model calibration, and this requirement for calibration may limit the routine use of the model. However, more information regarding ε may indicate some simple means of determining its value and so negate the need for calibration. Secondly, for the PEAK model to be widely used in assessing the effect of irrigation on soil salinity, it must be tested on soils and irrigation management data that could be readily collected by agricultural or environmental advisory agencies, rather than in controlled research studies. There may be uncertainties associated with such data which may limit application of the model. Finally, the model does not account for water flow through macropores and so its use in soils where macropore flow occurs, such as Vertisols, should give inaccurate predictions (Barry et al. 1985). However the degree of this inaccuracy has yet to be determined and so should be studied.

To provide more information on ε and to investigate possible limitations to the application of the PEAK model caused by input data uncertainties and macropore flow, the model was applied at eight sites using data which could be readily collected by agricultural or environmental advisory agencies. These sites were all located on commercial farms and covered a range in soil types common in irrigated areas of north-eastern Australia. All sites had previously been examined using mass balance approaches in a companion paper (Thorburn et al. 1990, subsequently referred to as Part I). In a third paper in the series (Thorburn and Rose 1990, subsequently referred to as Part III) a simple model for assessing the magnitude of macropore flow is developed.

Theory outline

The concentration of solute, at any depth in the soil profile and at any time resulting from a pulse application of solute (such as an irrigation event or fertiliser application) is calculated in two parts. Firstly, the depth of penetration of the solute concentration peak is calculated from water balance principles (Rose et al. 1982a). Secondly, dispersion around this peak is calculated using an analytical solution of the dispersion – convection equation (Rose et al. 1982b). Application of the theory assumes that; no significant soil-solute interactions or chemical transformations occur, the soil drains to a characterisable field capacity that is constant with depth, the profile is at field capacity below the root-zone, crop water uptake is uniform with depth to the bottom of the root-zone, and all water flows through the soil matrix. A brief description of the equations describing peak movement and dispersion are given here.

Peak movement (Rose et al. 1982a)

The depth, α (mm), of the solute concentration peak from the soil surface (downwards positive) in each event is calculated in finite time elements. For the n^{th} time element

$$\alpha_n = \alpha_{n-1} + \Delta W_{n,z=\alpha_{n-1}} / \theta_{fc} \quad (1)$$

where $\Delta W_{n,z=\alpha_{n-1}}$ is the equivalent ponded depth of water which passed below the previous position of the solute peak (α_{n-1}), and θ_{fc} (m³ m⁻³) is field capacity.

A general expression for $\Delta W_{n,z=\alpha_{n-1}}$ is

$$\Delta W_{n,z=\alpha_{n-1}} = I_n \Delta t - g(\theta_{fc} - \theta_{n-1}) + j E_n \Delta t / R_n \quad (2)$$

where

- Δt = length of the time element (chosen in this study to be a month),
- I = infiltration rate (mm month⁻¹)
- R_n = depth of rooting (mm),
- E = evapotranspiration rate (mm month⁻¹),
- g = depth over which the soil water deficit must be replenished for peak movement to occur (mm),
- j = proportion of evapotranspiration occurring before infiltration, and
- θ_{n-1} = volumetric moisture content at the end of the previous time increment (m³ m⁻³).

(Notation used in this paper are given in the Appendix.) $I \Delta t$ and $E \Delta t$ are expressed as equivalent ponded depths of water.

Lack of information concerning timing of water inputs within the calculation period leads to uncertainty in application of the model, but this uncertainty is bounded by two extreme possibilities; firstly that infiltration is complete before evapotranspiration, and secondly the reverse. Either of these two possibilities are chosen in Eq. (2) by selecting the value of the parameter j . If infiltration is assumed complete before evapotranspiration, then $j=1$, and for the reverse condition, $j=0$. If the value of j can not be chosen with confidence (e.g. from irrigation management data) calculations made using the extreme values (0 and 1) will encompass the uncertainty in the prediction of peak movement.

The value of g depends on the relationship between α_{n-1} and R_n . If $\alpha_{n-1} < R_n$, $g = \alpha_{n-1}$, but if $\alpha_{n-1} > R_n$, $g = R_n$. A knowledge of the previous profile water content is required to determine θ_{n-1} . θ_{n-1} is calculated from

$$\theta_{n-1} = \theta_{n-2} + (I_{n-1} - E_{n-1}) \Delta t / R_n \quad (3)$$

Values of θ will be between θ_{fc} and wilting point (θ_{wp}). If θ_{n-1} is calculated to be $> \theta_{fc}$ it is assumed that $\theta_{n-1} = \theta_{fc}$, and if θ_{n-1} is calculated to be $< \theta_{wp}$ it is assumed that $\theta_{n-1} = \theta_{wp}$.

Peak dispersion (Rose et al. 1982b)

The values of α from Eq. (1) provide a Lagrangian (moving) coordinate system about which dispersion takes place. Dispersion about the solute peak is calculated from the convection-dispersion equation for step changes in solute concentration under constant flux boundary conditions. It is assumed constant flux boundary conditions apply under non-steady conditions in the field (Wierenga 1977; de Smedt and Wierenga 1978). This solution can be modified by an approximate procedure to account for crop solute uptake (e.g. nitrate), not described here.

Solute is assumed to be applied as a rectangular pulse, of a given thickness (T) and initial concentration (C_0). Where solute is applied with irrigation water (or readily soluble fertilizer) Barry et al. (1985) showed that

$$T = I \Delta t / \theta_{fe} \quad (4)$$

If solute is applied in irrigation water, C_0 can be calculated from the solute concentration of the irrigation water and rainfall. The solutions for the leading edge (no solute – solute step) and trailing edge (solute – no solute step) of the rectangular solute pulse are superimposed (Rose 1977) to construct the resultant solute concentration, C (meq l^{-1}), from

$$C = C_0 \frac{1}{2} [\text{erfc}(A) - \text{erfc}(B)], \quad (5)$$

where

$$A = (z - \alpha) / (2 \sqrt{[D_0 t + \alpha \varepsilon]}),$$

$$B = (z - \beta) / (2 \sqrt{[D_0 t + \beta \varepsilon]}),$$

D_0 = molecular diffusion coefficient for the solute ($\text{mm}^2 \text{month}^{-1}$),

ε = coefficient of solute dispersivity in soil (mm),

z = soil depth (mm), and

$\beta = \alpha - T$.

The variable β is the depth of penetration of the trailing edge of the solute phase, which 'lags' behind the leading edge of the pulse, α , by the depth of the solute pulse, T .

Solute profiles at any time are simply calculated by repeating the calculation of C at successive soil depths. Where the solute considered is contained in irrigation water, solute is added to the soil with every irrigation. The movement and dispersion of each solute application is calculated, then superimposed (added) to give the resultant profile. The manner in which this was done is detailed below.

The value of ε , the only unknown in Eq. (5), can be adjusted to give agreement between model predictions and measured solute profiles. Once the value of ε is known for a site, future peak movement can be predicted.

Materials and methods

Site data

Eight sites were selected for this investigation from those examined in Part I, because they had the most detailed historical management data. Five sites were located in the Dee River Valley (DRV) and three in the Emerald Irrigation Area (EIA). (The location of these areas in Queensland was shown in Part I, Fig. 2.) The sites from the DRV had all been irrigated with groundwater, four being spray irrigated, and the fifth (site DHG-2) flood irrigated. (Sites are referred to using the same notation as in Part I.) The sites from the EIA were flood irrigated with dam water. Average annual irrigation plus rainfall infiltrating at the sites was between 760 and 2280 mm y^{-1} . Average class-A pan evaporation ranged from 1870 mm y^{-1} in the DRV to 2150 mm y^{-1} in the EIA. Soils comprised five cracking clay soils (Vertisols) and three non-cracking soils (Entisols and an Aridisols), ranging in texture from loam to heavy clay (Table 2). Irrigation water chloride concentration (C_w) and crops grown at each site are also shown in Table 2.

Table 2. Selected site details and the fitted value of ε for the soils studied

Site	Soil texture	Irrigation water chloride ^a (meq l^{-1})	Crop grown ^b	ε (mm)	F^c (%)
Dee River Valley (DRV)					
DCL	Loam	11.1	l	120	53.9
DHG-1	Loam	18.6	mb, w, b	100	19.4
DHG-2	Loam	18.8	mb, sf, b	20	4.8
DWD-1	Medium clay	14.5	l, s	10	5.6
DWD-2	Heavy clay	26.0	l	100	12.0
Emerald Irrigation Area (EIA)					
EAU-1	Medium clay	0.3	c	nr ^d	nr
EBU	Heavy clay	0.3	sb, c	nr	nr
ETB-1	Heavy clay	0.3	c, sf	nr	nr

^a long-term average

^b l = lucerne, s = sorghum, sf = sunflower, b = barley, c = cotton, sb = soybeans, w = wheat, mb = mung beans

^c leaching fraction = leaching flux/ I (from Part I)

^d nr = not reported (as explained in the text)

Soil sampling and analysis

Replicate soil cores were sampled at each site. Four cores were taken at DRV sites and 10 at EIA sites. Soils were analysed for chloride (Cl^-) at 100 mm depth increments. The three EIA sites had been sampled, approximately annually, for between 5 and 9 y. At these sites the final sampling was analysed because it provided the longest detailed management records. More complete details of soil sampling and analysis were given in Part I.

Irrigation, cropping and climatic data required in the PEAK model

Dee River Valley sites. For the DRV sites, cropping data, irrigation timing and amount of irrigation water applied were obtained from farmer records. C_w values were obtained from routine samplings (approximately bi-annual) conducted by the Queensland Water Resources Commission (unpublished data), and monthly values were interpolated from these data. Application rate of spray irrigation was approximately 15 mm h^{-1} at all sites.

Emeralds Irrigation Area sites. Crop planting, harvest and irrigation dates were recorded for the EIA sites. The amount of irrigation water infiltrating the soil was not directly determined, but was estimated as 80% of the plant available water capacity (PAWC) of the soils. This value was used because irrigations were scheduled at this soil water deficit, using the evaporation pan-crop factor model of Keefer et al. (1982). PAWC, θ_{fc} and θ_{wp} were predicted from -1.5 MPa soil water contents using the relationships of Shaw and Yule (1978), as was described in Part I. The 'field capacity concept', that soil water contents decrease over 2 to 3 days after complete profile wetting to a steady value (Veihmeyer and Henderickson 1949), applies in these soils (Thorburn and Gardner 1986; Yule unpublished data). C_w values were taken from Shaw and Yule (1978).

Crop data. The variation in the effective rooting depth (R_n) with time was based on data from local agronomists and extension officers. For annual crops R_n in the first month after planting was taken to be 60% of the maximum, and subsequently equal to the maximum value. For all crops except mung beans, the maximum value of R_n was assumed equal to the depth of maximum soil Cl^- in soils

with steady state root-zone Cl^- (the steady-state soil Cl^- data being taken from calculations in Part I). This method of estimating maximum R_n has been found accurate in many irrigated soils in Queensland (Gardner et al. 1984; Thorburn and Gardner 1986; Baker and Ahern 1989), and values used agreed with local experience (Palu 1987; Bartholomew 1984). For mung beans the maximum value of R_n was taken as 600 mm (Palu 1987). At sites where lucerne was grown continuously, short-term variations in R_n due to crop re-establishment (every 2 to 3 y) were not considered to have a significant long-term effect on solute movement and so were ignored. During fallow periods R_n was taken as 200 mm to account for soil evaporation, as was done by Rose et al. (1982c).

Crop Cl^- uptake can be typically expected to be <10% of that applied with irrigation (Oster 1984). This value is in the range of uncertainty in field applications of solute transport models (Rose et al. 1982c), and so crop Cl^- uptake was neglected. Irrigation and rainfall were the only sources of Cl^- at all sites.

Evapotranspiration data. Evapotranspiration (E) was estimated as a proportion of monthly pan evaporation. These proportions, or crop factors, were given for cotton by Keefer et al. (1982), and similar values were used for grain crops. For lucerne the crop factor was assumed constant and equal to 80% of pan evaporation (Bartholomew 1984). Short-term variations in E due to lucerne stand re-establishment were not considered significant, and so were not accounted for. Crop factors for mung beans were given by Palu (1987).

Pan evaporation (and rainfall) values were taken from the nearest weather recording stations. These stations were within 3 km of the sites.

Application of theory to the data

All calculations used the BASIC computer programs of Thorburn et al. (1987b) with a monthly time step. This time step was chosen as suitable for the level of detail of the data. The amount of irrigation applied at each site in each month was added to rainfall to give $I_n \Delta t$ (monthly infiltration). Runoff of rainfall was assumed negligible and not deducted from rainfall when calculating $I_n \Delta t$. For each month C_0 was calculated as the mean of C_w and rainfall Cl^- concentration, weighted for the different amounts of rain and irrigation as described in Part I. In all simulations the value of D_0 was taken as $100 \text{ mm}^2 \text{ day}^{-1}$, as was done by Rose et al. (1982a–c) and Barry et al. (1985).

Each infiltration event not only adds a pulse of Cl^- to the soil, but displaced Cl^- concentration peaks from previous irrigations down the profile. Therefore the effect of previous irrigations on the soil Cl^- concentration had to be accounted for in simulating Cl^- profiles at the time of sampling (i.e. time n). This was done by calculating the separate Cl^- profiles at time n , that resulted from the n^{th} irrigation and all previous irrigations (i.e. $n-1$, $n-2$, etc.) as illustrated in Fig. 1. These Cl^- profiles were then added, employing the principle of superposition (Rose 1977), to give the simulated profile at time n (Fig. 1). Current Cl^- profiles arising from increasingly 'old' irrigations were calculated until the depth of Cl^- penetration into the soil was great enough that there was a negligible contribution to solute in the root-zone at the time of sampling. This negligible contribution was judged to occur when simulated values of C within the root-zone, caused by an irrigation at a particular time before time n , were $<0.1 \text{ meq l}^{-1}$. This arbitrarily chosen value is well below the accuracy of soil Cl^- determinations (1 meq l^{-1}). Initial solute profiles were also assumed to be displaced downwards, and so not contribute to solute within the root-zone at time of sampling.

Values of C were calculated at 100 mm depth increments, commencing at 50 mm soil depth (to align simulated C values with the centre of the depth interval over which soil samples were taken) to give profiles of the depth distribution of solute. Values of j (Eq. (2)) and ε (Eq. (5)) were varied to fit simulated to measured data. At some

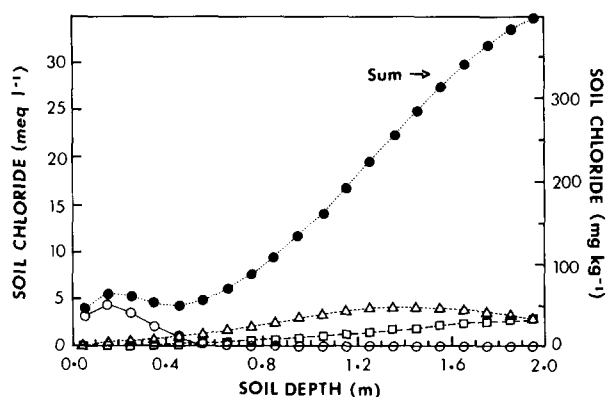


Fig. 1. Simulated soil chloride profiles in the soil at time n (dashed lines), arising from irrigations at time n ($\circ$), and 8 (Δ) and 12 ($\square$) months prior to time n . All profiles arising from previous irrigations (including others not shown) are summed to give the total soil chloride profile at time n (solid symbol)

DRV sites assumptions were made regarding uncertainties in input data, and these are described below.

Model parameters ε and j were varied to give agreement between measured and simulated solute profiles at each site. However at many sites there was uncertainty regarding some input data, as is common in practical field situations. The effect of this uncertainty on simulations was also examined, to indicate both the real difficulties in practical application of models and the sensitivity of simulations to variability in input data.

Results

Soil Cl^- values calculated at each 100 mm depth increment for all eight sites are plotted against measured values in Fig. 2. Generally, measured profiles could be accurately predicted at the DRV sites, but were underestimated at the EIA sites.

Dee River Valley sites

Values of ε chosen to give a good fit between simulated and measured profiles varied from 10 to 120 mm (Table 2), a range similar to that found in previous studies (Table 1). Values of ε were not related to soil type, soil texture or management variables. However, ε values were significantly correlated ($r=0.90$) with the logarithm of the leaching fraction (leaching fractions were calculated in Part I and are given in Table 2) and the logarithm of the leaching flux ($r=0.87$). The regression of leaching fraction (F , %) and ε (mm) was

$$\varepsilon = 46.6 \ln(F) - 48.5 \quad (n=5). \quad (6)$$

It is not suggested that this regression is a universal description of the relationship between F and ε . It is given to allow comparison with other data.

At all sites a value of $j=1$ (indicating evapotranspiration occurred before irrigation) was found to give the closest fit of simulated to measured Cl^- profiles. This value of j is consistent with the practice of irrigating to replenish soil water deficits, as is normal in the DRV (and EIA). Variation in j over the possible range (0 or 1) had a

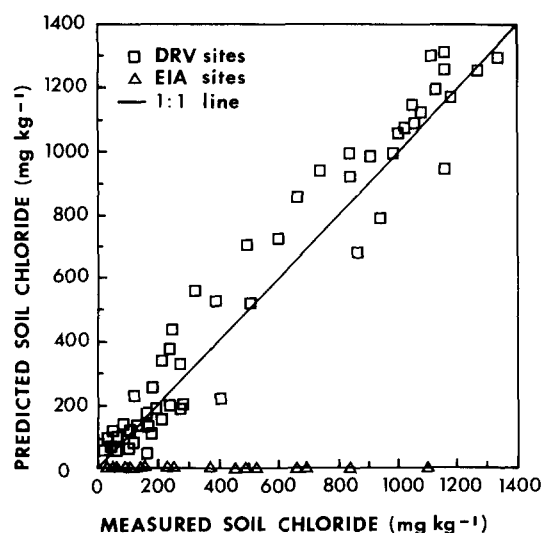


Fig. 2. Comparison between soil chloride measured or predicted by the PEAK model for five sites from the Dee River Valley (DRV) and three from the Emerald Irrigation Area (EIA). Each point represents one depth increment in a soil profile

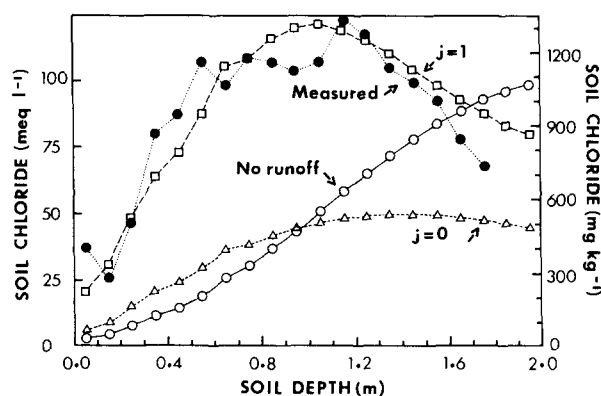


Fig. 3. Soil chloride profiles measured and simulated by the PEAK model at site DWD-2. The various simulated profiles were calculated assuming evaporation before transpiration ($j=1$); evaporation after transpiration ($j=0$); and that all water infiltrated the soil matrix (i.e. no runoff) with $j=1$, as marked on the figure

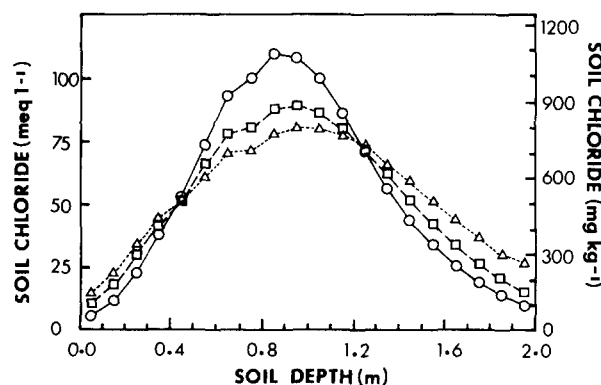


Fig. 4. Soil chloride profiles simulated by the PEAK model at site DWD-1 using different values of the solute dispersivity coefficient (ϵ): $\epsilon=10$ mm ($\circ$), $\epsilon=70$ mm ($\square$), and $\epsilon=120$ mm (Δ). $\epsilon=10$ mm corresponds to the 'spring planting' simulation of Fig. 5

greater effect on model predictions (Figure 3) than did large variations in ϵ (Fig. 4).

At sites DWD-1 and DWD-2, if infiltration was considered equal to the sum of irrigation and rainfall in all months, predicted soil Cl^- values were considerably lower than measured values (e.g. Fig. 3). Irrigation application rates (15 mm h^{-1}) would have been less than the steady-state infiltration rate of these soils ($> 25 \text{ mm h}^{-1}$, Thorburn et al. 1987a). Thus under irrigation, surface ponding was unlikely and bypass flow (Scotter 1978) or runoff (Rubin 1966) would not have occurred. Only under periods of heavy rainfall was it likely that ponding would have occurred on these two Vertisols. $I_n \Delta t$ was thus reduced in months when heavy rain (e.g. 150 to 330 mm) had fallen over two or three days. During these months, $I_n \Delta t$ was calculated as 50% of rainfall, giving predictions close to measured soil Cl^- profiles at these sites (Figs. 3 and 5). The value of 50% was chosen by 'trial and error', and resulted in an average reduction in $I_n \Delta t$ of 142 mm y^{-1} over the period of analysis. Varying the proportion of rainfall included in calculation of $I_n \Delta t$ by $\pm 10\%$ of rainfall did not have a marked effect on predictions. The implications of this reduction in $I_n \Delta t$ is discussed below.

Lucerne was planted during the year prior to sampling at site DWD-1 after six years of grain sorghum. However, the actual month when lucerne was planted at this site is not known. The two likely planting times would have been autumn (April) or spring (September) in the year prior to sampling. The good agreement between the simulated spring planting and measured soil Cl^- profiles (Fig. 5) indicated this to be the most likely planting time.

At all sites, there was uncertainty in the amount of water applied at each irrigation. This uncertainty was greatest in relation to irrigation application at sites DHG-1 and 2, where between 25 and 35 mm of water was applied with each irrigation. Predictions made using the minimum and maximum possible irrigation inputs at site DHG-1 differed, although the differences were minor. The minimum value gave predictions closest to the measured soil Cl^- profile at this site, and this value then used in calculations for site DHG-2.

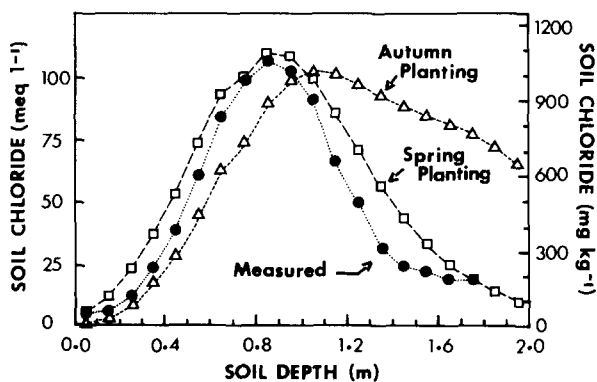


Fig. 5. Soil chloride profiles measured and simulated by the PEAK model at site DWD-1. The two simulated profiles were calculated assuming different planting times of lucerne, following grain sorghum and a fallow period of uncertain length

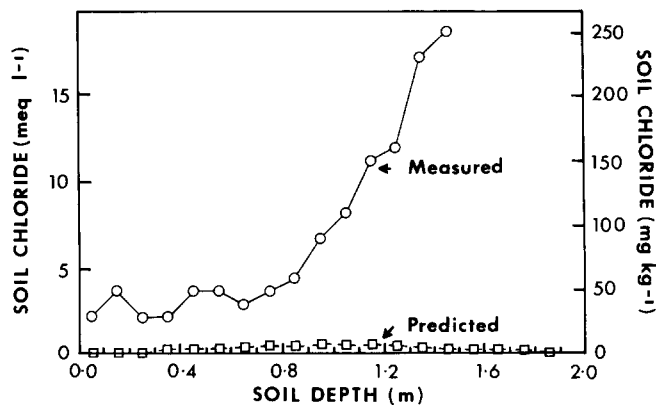


Fig. 6. Soil chloride profiles measured and predicted by the PEAK model at site EBU

Emerald Irrigation Area sites

At the EIA sites predicted soil Cl^- values were consistently lower than measured values (Figs. 2 and 6) regardless of the value of ε (over the range 10 to 500 mm) of j (either 1 and 0) used. Varying $I_n \Delta t \pm 20\%$ (to account for uncertainty in its calculation) had little effect on predictions. Taking $j = 1$ gave higher predictions compared with $j = 0$, although these values were still much lower than those measured. Variation of j had greater effect on predictions than did ε .

Discussion

Dee River Valley sites

The fact that the PEAK model could provide good predictions of soil Cl^- profiles over the range of soils studied in the DRV using historical management data from farmer records (Figs. 2, 3 and 5) indicates that the model has scope for wider application with this type of input data. Where there was uncertainty in management data simulations could be used to investigate this uncertainty and indicate likely values for inputs.

Emerald Irrigation Area sites

The underestimation of soil Cl^- levels at all EIA sites (Figs. 2 and 6) indicates that assumptions made in the model were not met in the field at these sites. Assumptions not met at the EIA sites were that all water flowed through the soil matrix (discussed in the following section) and that θ_{fc} and θ_{wp} values remained constant with depth (Table 3). Also, a watertable had formed in the rootzone of site EBU in response to irrigation. Water and solutes moved upward from this watertable increasing root-zone Cl^- levels (Part I), a process not considered in the PEAK model.

To test the effect that depth variations in θ_{fc} and θ_{wp} had on predictions the PEAK model was modified to account for this variation. In this modified procedure the

Table 3. Variation in field capacity (θ_{fc}) and wilting point (θ_{wp}) with soil depth at site EAU-1

Depth (m)	θ_{fc} ($\text{m}^3 \text{m}^{-3}$)	θ_{wp} ($\text{m}^3 \text{m}^{-3}$)
0.1	0.45	0.26
0.2	0.45	0.28
0.3	0.45	0.31
0.4 ^a	0.45	0.31
0.5	0.42	0.33
0.6	0.42	0.33
0.7 ^a	0.42	0.33
0.8	0.41	0.36
0.9	0.41	0.36
1.0 ^a	0.41	0.36
1.1	0.39	0.38
1.2 ^a	0.39	0.38

^a Bottom depth of soil 'section' used in the layered PEAK model (as explained in the text)

root-zone was divided into an arbitrarily defined number of sections (e.g. four, Table 3). Values of θ_{fc} and θ_{wp} were assumed constant within each root-zone section, but could vary between sections. Peak movement was calculated (from Eqs. (1) to (3)) sequentially through each section, starting with the surface section. If α_n was deeper than the lower boundary of the layer (Z_b), $\Delta W_{n,z=Z_b}$ was calculated and taken as the infiltration into the section below. θ_n was also updated for each section separately. Accounting for depth variations in θ_{fc} and θ_{wp} only increased predicted soil Cl^- values by $<0.2 \text{ meq l}^{-1}$, so providing no significant improvement in prediction at these sites. This lack of response to depth variations in θ_{fc} and θ_{wp} occurred because the usual values of $I_n \Delta t$ and $E_n \Delta t$ were such that there was either no peak movement (i.e. $\theta_n < \theta_{fc}$) or $I_n \Delta t$ was great enough to move the solute peak well below the root-zone (e.g. to 2 or 3 m soil depth). Under conditions where infiltration and evapotranspiration events are less extreme and solute peaks are within the root-zone for longer periods variation in θ_{fc} and θ_{wp} with soil depth may be of more importance in calculating peak movement.

Although the PEAK model did not represent the solute transport processes occurring in the EIA sites, other more complex models are available which describe these processes (e.g. those of Rao et al. 1980; Skopp et al. 1981 or van Genuchten and Dalton 1986). Also upward movement of solutes from a high watertable can be calculated from soil hydraulic properties and evaporation data (Talsma 1963). These models may not be suitable for routine application by agricultural or environmental advisory agencies due to practical limitations in determining required input data. Alternatively, the simpler dynamic mass balance model of Rose et al. (1979) provided a good description of depth averaged soil Cl^- at these sites (Part I), and could be used to assess the longer-term effect of irrigation on salinity of these soils.

Effect of soil cracks on model predictions

Rose et al. (1982b and 1983) and Barry et al. (1985) noted that the PEAK model should not give accurate predictions of solute movement in soils where water moved through cracks or pores and bypassed the soil matrix. Alternatively, bypass flow may lead to large apparent values of ε because of the increase in pore water velocity (Dyson and White 1987) and depth of solute penetration. The five Vertisols studied were strongly cracking and so the PEAK model should not have given accurate predictions in these soils.

In the two DRV Vertisols (sites DWD-1 and DWD-2) the PEAK model was able to reproduce measured soil Cl^- profiles (Figs. 3 and 5) providing allowance was made for runoff and/or bypass flow of rainfall. As these sites were flat runoff would have been minimal, and rainfall that did not infiltrate the soil matrix would have flown through macropores. This macropore flow must have not contributed to Cl^- movement in these soils as ε values were not large in comparison to those determined in non-cracking soils in this (Table 2) or previous studies (Table 1). An objective means of determining the reduction in $I_n \Delta t$ required to account for bypass flow (or runoff if necessary) would be preferable to the subjective method used here. A method of estimating bypass flow is developed in Part III, which may be useful for this purpose.

In the EIA soils the assumption of uniform water flow through the soil matrix was not valid, as irrigation water is known to infiltrate dominantly through cracks in these soils (Shaw and Yule 1978). No successful adjustment could be made to the PEAK model to account for infiltration into cracks, as was done for the DRV sites. The higher 'resident' soil Cl^- compared with predicted soil Cl^- indicates that solutes were not being efficiently leached from these soils (Thomas and Philips 1979; Rose et al. 1983). Soil Cl^- levels were decreasing slowly with time at sites EAU-1 and ETB-1 (Part I), due to diffusion of Cl^- from soil aggregates into soil water at high soil water contents and then convective movement of Cl^- downward in the profile. These processes are not described by the PEAK model but are considered in more complex models, as noted above.

Determination of ε and j

Values of ε have been related to pore size distribution and aggregate size, but should be independent of pore water velocity (Jensen 1983). Shaw and Thorburn (1985) found that leaching fraction was well correlated with soil properties (exchangeable sodium, clay and cation exchange capacity). They hypothesised that those soil properties influenced soil structure and hence soil permeability. The correlation between ε and leaching fraction for the DRV sites (Eq. (6)) thus implies that similar soil structural factors may influence both solute dispersivity and leaching in soils. This is physically plausible. The correlation derived in this study was based on too few points to be of any real predictive value. However, it indicates an avenue for future investigation.

As values of ε can not be accurately predicted from soil or irrigation variables, values of ε should be determined by model calibration before application of the PEAK model at a particular site. Where such model calibration is not possible however, using the values 10 and 120 mm for ε in the simulation of soil Cl^- profiles appears likely to indicate the range of variation of ε for Cl^- profiles in the root-zone of field soils. Whether ε is more likely to be at the higher or lower extreme of this range at a site is indicated by the leaching fraction at the site (Eq. (6)). Alternatively, since this apparently extreme range ε values usually gave relatively small differences in simulated soil Cl^- profiles (Figure 4), the adoption of a mean value (such as 70 mm) may be adequate for management purposes, even if results are approximate. Certainly variation in ε (Fig. 4) generally had less effect on predictions than the value chosen for j (Fig. 3), or lack of information on irrigation management or agronomic variables (Fig. 5).

Although the assumed value of j had a large effect on simulated soil Cl^- values, its value can be deduced from knowledge of irrigation practice, as was done by Barry et al. (1985) and in the sites studied in this paper.

Considerations for future application of the PEAK model

Our results, and those of previous studies, have shown that the PEAK model is applicable over a range of soil types and irrigation practices, and due to its simplicity, could be widely used to predict non-reactive solute movement in field soils. However, the model may not be applicable in all situations because of unsatisfied assumptions. Consideration of the conditions under which the model has already been tested (in this and other studies) will help define the future conditions under which the model will be likely to prove useful.

For the light textured soils, the PEAK model has been found to provide good predictions of Cl^- and nitrate movement and further application of the model in these soils is justified. The assumptions in the model are likely to be met in these soils, and predictions will be sufficiently accurate to be useful in making irrigation management decisions.

For the Vertisols studied, the PEAK model gave accurate predictions of Cl^- profiles in soils which were spray irrigated. In contrast, where Vertisols were flood irrigated model predictions were inaccurate because of preferential irrigation water movement or the occurrence of high water-tables. Thus the decision whether to use the PEAK model on Vertisols should be based on factors that affect preferential water movement or the occurrence of high watertables. Such factors would be either a comparison of soil matrix hydraulic conductivity and likely irrigation application rates, to assess the possible importance of preferential soil water flow in solute movement (Beven and Germann 1982); or landscape conditions (e.g. soil depth and relative soil permeabilities), to assess the probability of high watertables forming in irrigation areas (Shaw et al. 1987).

Conclusions

The PEAK model gave good predictions of measured soil Cl^- values and profiles in a range of field soils on commercial farms, including spray irrigated Vertisols. In Vertisols that had been flood irrigated, model predictions of soil Cl^- were lower than measured data due to infiltration of water into cracks, and at one site, the effects of upward flow from high watertables. Both these processes are not represented in the model.

Management data from detailed farmer records proved adequate for application of the PEAK model in most cases. Where there was uncertainty in management data, the effect of the uncertainty in input parameters on soil Cl^- profiles was able to be studied. The parameter j could be deduced from management practice, and uncertainty in parameter ε did not have a large effect on predictions in practical terms. Application of the model also provided useful insights into the root-zone hydrology of the soils studied. Generally, the model was found to be well suited to field application for interpreting soil solute dynamics over a wide range of soil types.

Appendix. List of notation used in the paper

α	= depth of solute concentration peak from the soil surface (mm),
β	= $\alpha - T$,
ε	= coefficient of solute dispersivity in soil (mm),
Δt	= length of the time element (taken as one month),
θ	= volumetric moisture content ($\text{m}^3 \text{m}^{-3}$),
θ_{fc}	= field capacity ($\text{m}^3 \text{m}^{-3}$),
θ_{wp}	= wilting point ($\text{m}^3 \text{m}^{-3}$),
$\Delta W_{n, z = \alpha n - 1}$	= equivalent ponded depth of water which passed below the previous position of the solute peak, α_{n-1} (mm),
C	= soil solute concentration,
C_0	= initial solute concentration (of infiltrating water) (meq l^{-1}),
C_w	= solute concentration of irrigation water (meq l^{-1}),
Cl^-	= soil chloride measured in 1:5 soil:water suspension (mg kg^{-1}),
D_0	= molecular diffusion coefficient for the solute ($\text{mm}^2 \text{month}^{-1}$),
E	= evapotranspiration rate (mm month^{-1}),
F	= leaching fraction [= leaching flux/ I (from Part I)],
g	= depth over which the soil water deficit must be replenished for peak movement to occur (mm),
I	= infiltration rate of irrigation + rain (mm month^{-1}),
j	= proportion of evapotranspiration occurring before infiltration,
n	= number of time element,
R	= depth of rooting (mm),
T	= thickness of a rectangular pulse of solute [where solute is applied with irrigation water or as readily soluble fertilizer, $T = I \Delta t / \theta_{fc}$],
z	= soil depth (mm), downward positive,
Z_b	= depth of base of soil layer (mm).

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